

Sports Physiology

It is the study of long and short-term effects of training and body conditions of athletes.

Physiological Factors Determining Components of Physical Fitness

Strength

- Size of muscles
- Body weight
- Muscle composition
- Intensity of nervous impulse

Speed

- Mobility of nervous system
- Muscle composition
- Explosive strength
- Flexibility

Endurance

- Aerobic capacity
- Lactic acid tolerance
- Movement economy
- Muscle composition

Flexibility

- Muscle strength
- Age and gender
- Stretchability of muscle
- Internal environment
- Previous injury

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Effects of Exercise on Cardiorespiratory System

- Increase in the size of the heart
- Decrease in resting heart rate
- Increases blood flow
- Decrease in blood pressure
- Increase in blood volume
- Decrease in rate of respiration
- Increase in endurance
- Increase in lung efficiency

Effects of Exercise on Muscular System

- Change in shape and size of muscles
- Formation of more capillaries
- Control extra fat
- Increases food storage
- Exercise and its health benefits
- Non-functioning fibers become active
- Efficiency and movement of muscles
- Body posture remains correct

Sports Injuries

It is common in sports. During practice, training, or competition, any player can be injured. It is natural to get injured.

Classification of Sports Injuries

- Soft tissue injuries
- Bone injuries
- Joint injuries

Soft Tissue Injuries

Contusion

Direct impact or impact without any sports equipment can be the main cause of it. It is due to a minor accident.

Strain

It can be mild or severe. It can occur at any time during practice or competition. It is caused by twisting or pulling of muscles.

Sprain

It is an alignment injury. It may occur due to overstretching or tearing of ligaments.

Abrasion

It is a major injury that usually occurs due to friction with certain equipment or falling over an area where the bones are very close to the skin.

Laceration

A laceration is a kind of tear-like wound or cut within the skin or flesh that is open and may be deep, caused by a blunt impact to soft tissue. It is an injury in the form of a cut owing to the separation of connective tissues.

Incision

A clean, straight cut caused by a sharp edge (i.e., a knife). Tends to bleed heavily as multiple vessels may be cut directly across. Connecting structures such as ligaments and tendons may also be involved.

Bone Injuries

Simple Fracture

Fracture of the bone only, without damage to the surrounding tissues or breaking of the skin.

Compound Fracture

A compound fracture is a break or crack in your bone that is visible through your skin.

Transverse Fracture

It is a fracture where the bone breaks at a right angle to the long axis of the bone. Transverse fractures most often occur as the result of strong force applied perpendicular to the long axis of a bone.

Oblique Fracture

An oblique fracture is characterized by a break that is curved or at a diagonal angle to the bone. A sharp blow that comes from an angle (i.e., above or below) may cause oblique fractures.

Complicated Fracture

Complicated fracture – structures surrounding the fracture are injured. There may be damage to the veins, arteries, or nerves.

Greenstick Fracture

A greenstick fracture is a fracture in a young, soft bone in which the bone bends and breaks.

Impact Fracture

An impacted fracture occurs when the broken ends of the bone are jammed together by the force of the injury.

Joint Injuries

Dislocation of joints is a main injury. Players usually face dislocation of joints in their fingers and hands.

- Dislocation of lower jaw
- Dislocation of shoulder joint
- Dislocation of hip joint
- Dislocation of wrist

